

Name: _____ Date: _____

Instructions. Show all work with clear, logical steps. No work or hard-to-follow work will lose points. Scientific calculators are allowed.

Problem 1. (4 points)

Find the partial fraction decomposition of $\frac{108}{x^2 - 81}$.

(NO NEED TO EVALUATE THE INTEGRAL!)

Problem 2. (4 points) Use partial fractions to evaluate the integral

$$\int \frac{5x + 16}{x^2 + 8x} dx.$$

Remember—writing your name and full date at the top earns 2 free points!